Variant Calling EGFR gene

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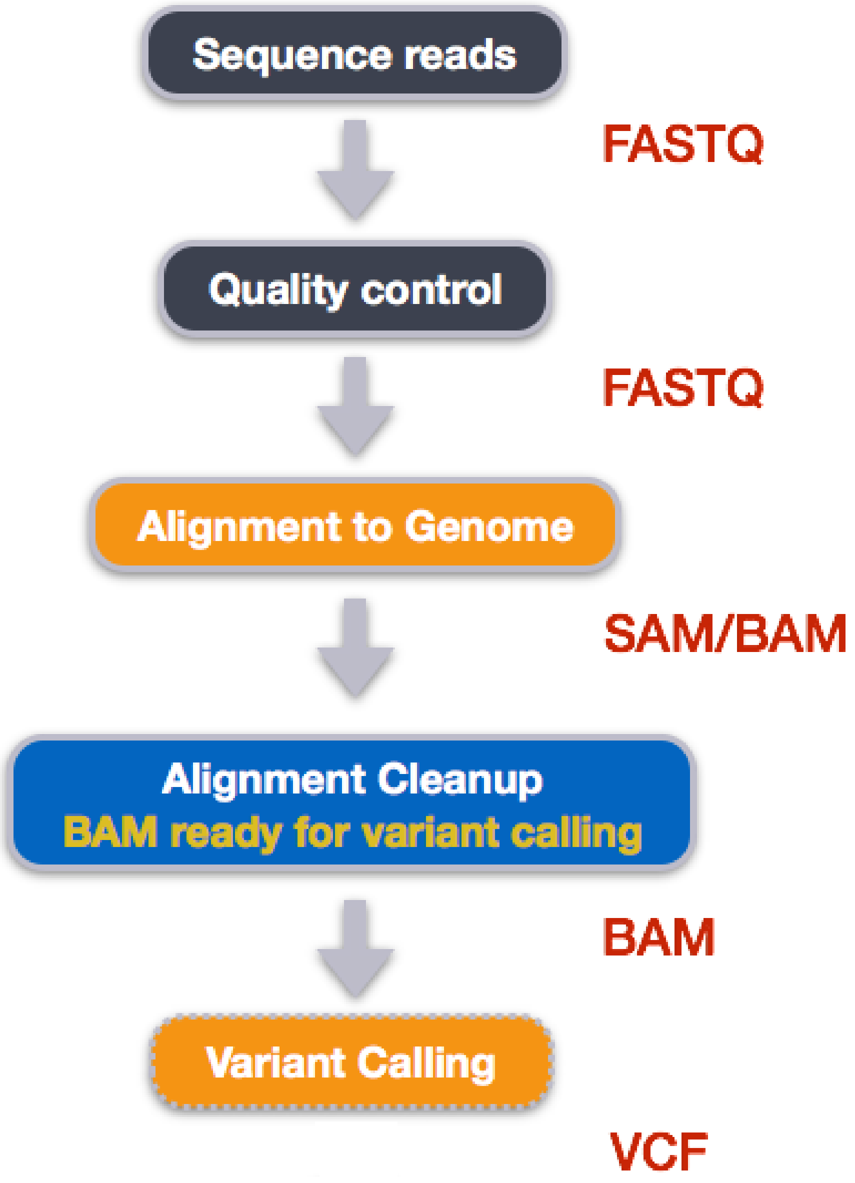
Variant Calling Workflow

FASTQ → BAM → coverage → VCF → biological interpretation

Variant calling compares sequencing reads to a reference genome to detect genetic changes, allowing us to identify candidate mutations and study their biological relevance.



FastQ structure

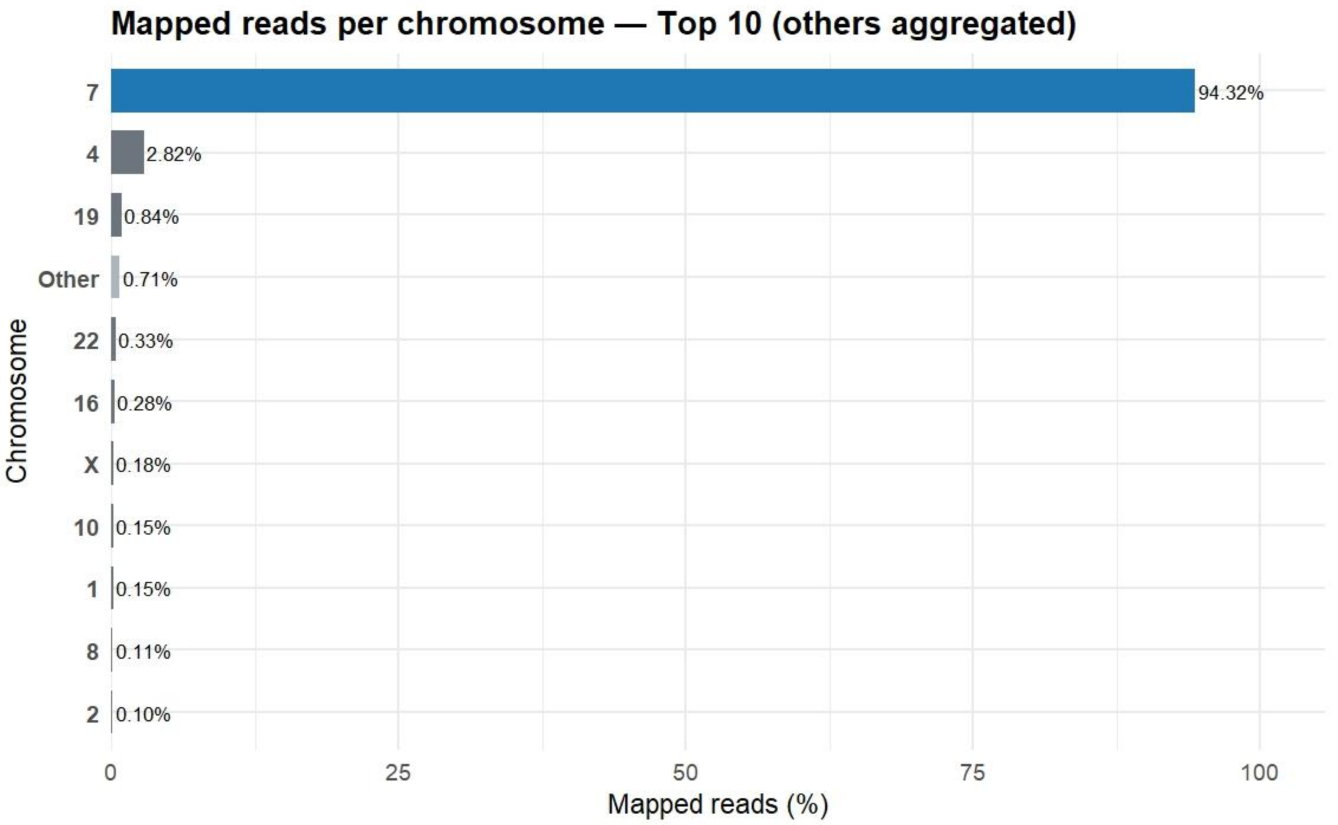
- No adapter sequences detected in the reads, indicating prior adapter trimming.
- FASTQ files (aln1/aln2) contain primer sequences followed by partial amplicon sequence.
- Mean read length: **151 bp**.

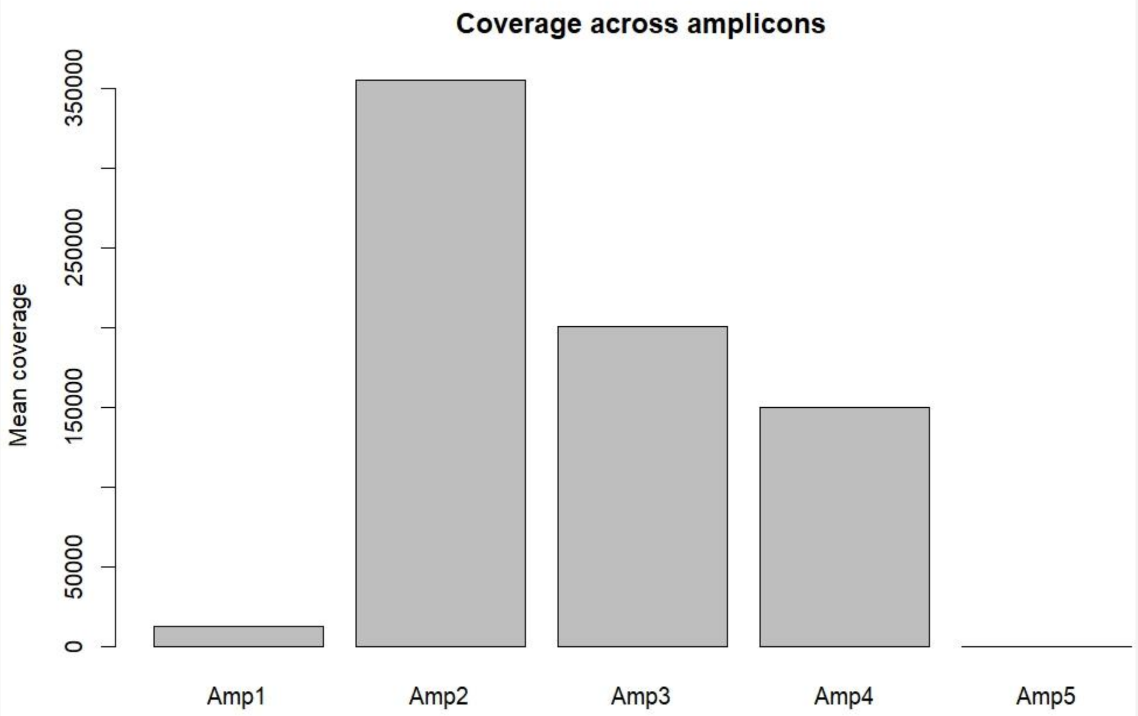
```
> sread(aln1)[1:10]
DNAStringSet object of length 10:
     width seq
 [1]   151 TTGCCAGTTAACGTCTTCCTTCTCTCTCTGTCATAGGGACTCTGGATCCCAGAAGGTG...AGAGAAGCAACATCTCCGAAAGCCAACAAGGAAATCCTCGATGTGAGTTTCTGCTTTG
 [2]   151 CACACTGACGTGCCTCTCCCTCCCTCCAGGAAGCCTACGTGATGGCCAGCGTGGACAA...CCTCCACCGTGCAACTCATCACGCGGCTCATGCCCTTCGGCTGCCTCCTGGACTATGT
 [3]   151 TGGCCAGTTAACGTCTTCCTTCTCTCTCTGTCATAGGGACTCTGGATCCCAGAAGGTG...CCGAAAGCCAACAAGGAAATCCTCGATGTGAGTTTCTGCTTTGCTGTGTGGGGGTCCA
 [4]   151 TTGCCAGTTAACGTCTTCCTTCTCTCTCTGTTATAGGGACTCTGGATCACAGAAGGTG...AGAGAAGCAACATCTCCGAAAGCCAACAAGGAAATCATCGATGAGAGTTTCTGCTTTG
 [5]   151 TTGCCAGTTAACGTCTTCCTTCTCTCTCTGTCATAGGGACTCTGGATCCCAGAAGGTG...AGAGAAGCAACATCTCCGAAAGCCAACAAGGAAATCCTCGATGTGAGTTTCTGCTTTG
 [6]   151 TGGCCAGTTAACGTCTTCCTTCTCTCTCTGTCATAGGGACTCTGGATCCCAGAAGGTG...AGAGAAGCAACATCTCCGAAAGCCAACAAGGAAATCCTCGATGTGAGTTTCTGCTTTG
 [7]   151 CACACTGACGTGCCTCTCCCTCCCTCCAGGAAGCCTACGTGATGGCCAGCGTGGACAA...CCTCCACCGTGCAGCTCATCACGCAGCTCATGCCCTTCGGCTGCCTCCTGGACTATGT
 [8]   151 TTGCCAGTTAACGTCTTCCTTCTCTCTCTGTCATAGGGACTCTGGATCCCAGAAGGTG...AGAGAAGCAACATCTCCGAAAGCCAACAAGGAAATCCTCGATGTGAGTTTCTGCTTTG
 [9]   151 TTGCCAGTTAACGTCTTCCTTCTCTCTCTGTCATAGAGACTCTGGATCCCAGAAGGTG...CCGAAAGCCAACAAGGAAATCCTCGATGTGAGTTTCTGCTTTGCTGTGTGGGGGTCCA
[10]   151 CACACTGACGTGCCTCTCCCTCCCTCCAGGAAGCCTACGTGATGGCCAGCGTGGACAA...CCTCCACCGTGCAACTCATCACGCAGCTCATGCCCTTCGGCTGCCTCCTGGACTATGT
>
```

Primer	Reads	Reads with 1 mismatch
F1	203 160	216 665
F2	6 432	8 452
F3	97 037	105 619
F4	144 222	153 206
	Tot: 450 851 78.4 %	Tot: 483 942 84.1%
R1	182 673	207 288
R2	8 056	8 871
R3	90 461	103 928
R4	128 542	148 225
	Tot: 409 732 71.3%	Tot: 468 312 81.4%

Coverage

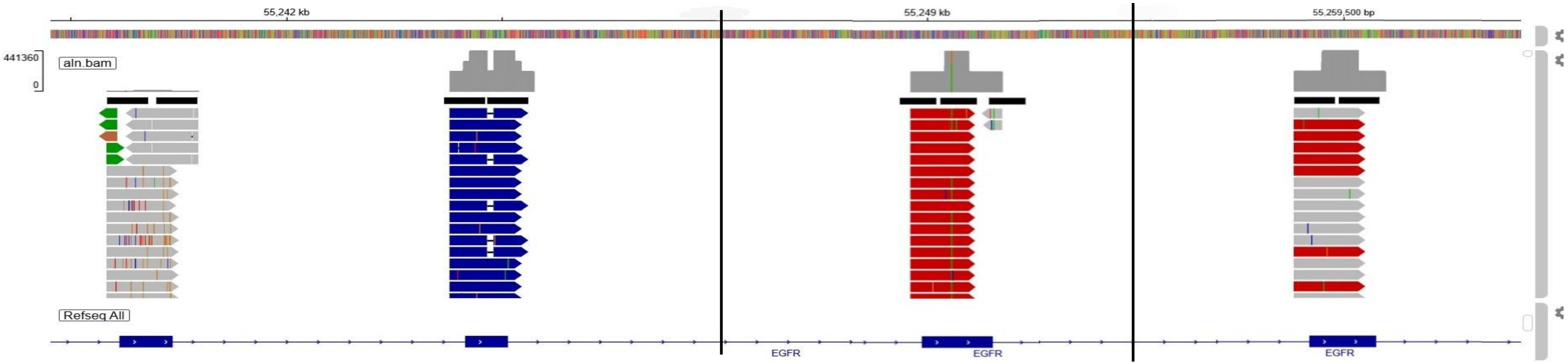
- Chromosome 7 contains **~94%** of the mapped reads, indicating a highly targeted experiment.
- Five enriched regions (coverage peaks) were detected on chr7; four true EGFR peaks, one off-target peak.





Four peaks cluster at the **EGFR locus** and represent the intended amplicons;
The fifth peak is weakly supported and likely an off-target.

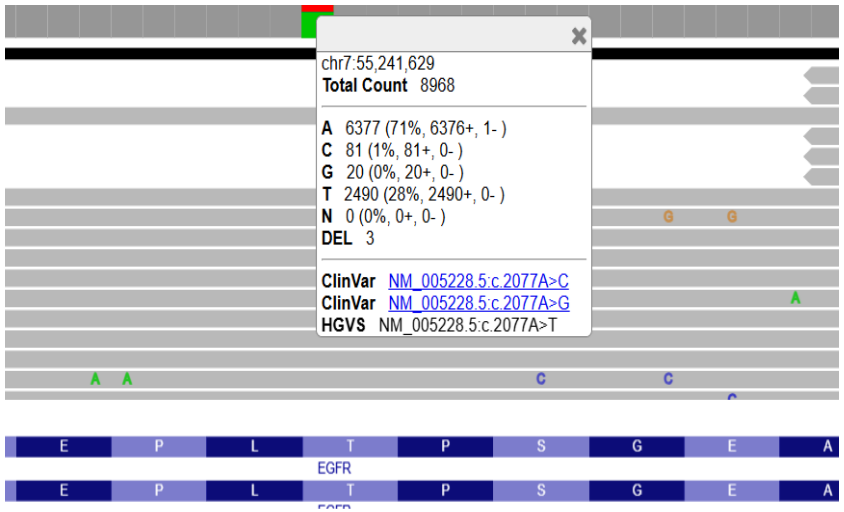
Amplicon	Coord	Mean Coverage	Interpretation
Amp1	chr7:55,241,584–55,241,796	12 654	Underrepresented; GC-rich possible bias
Amp2	chr7:55,242,379–55,242,574	355 169	Very high coverage
Amp3	chr7:55,248,957–55,249,194	200 532	High coverage
Amp4	chr7:55,259,375–55,259,589	150 305	High coverage
Amp5	chr7:129,945,985–129,946,111	178	Likely off-target / noise



Variant calling

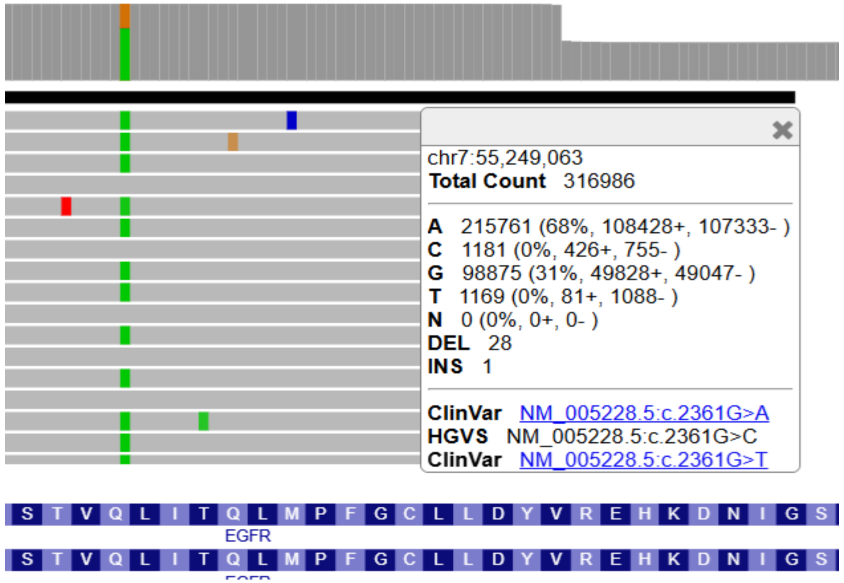
Performed using iVar on the Galaxy platform

Minimum quality score threshold to count base = 20
Minimum frequency threshold = 0.15

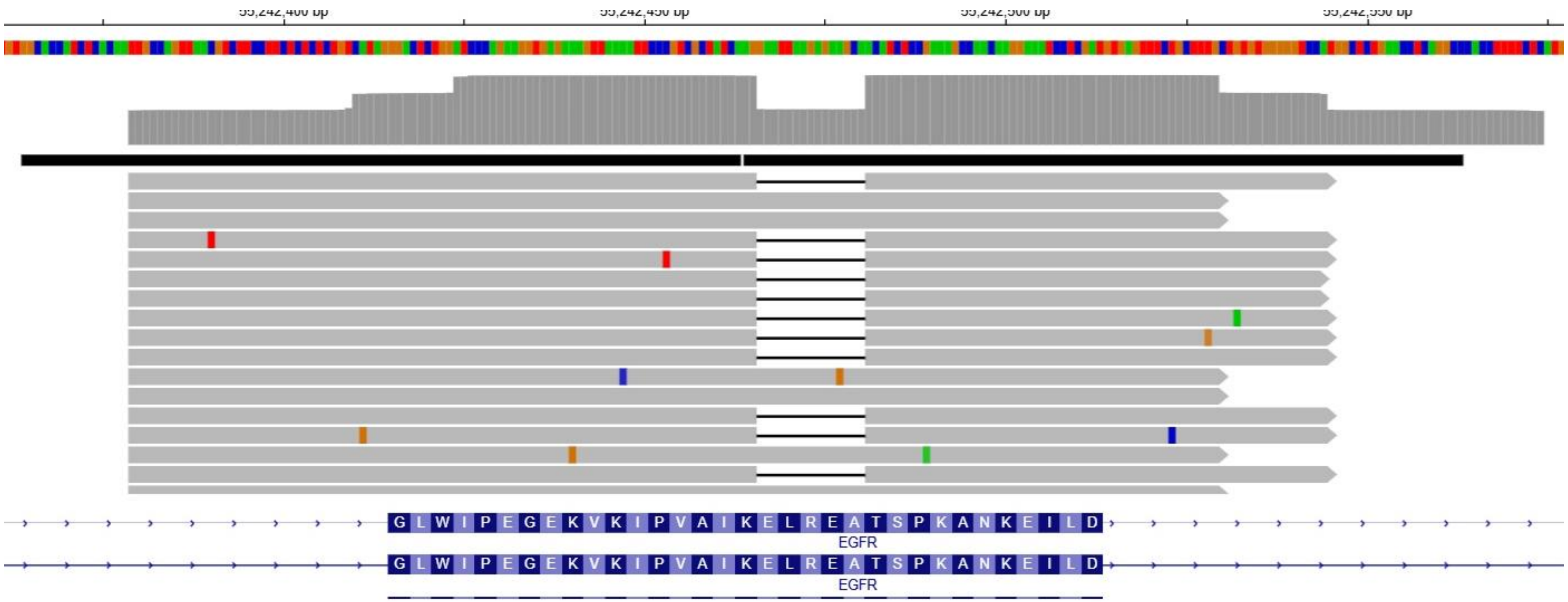


Chr7:55,241,629
Amp1
Missense
A>T

Variant	Genomic Position	Effect	Variant type	AF %	Total depth	ALT reads	Confidence
1	chr7:55,241,629	Missense	A>T	28	8 968	2 490	Medium
2	chr7:55,249,063	Synonymous	A>G	31	316 986	98 875	High
3	chr7:55,242,465	Inframe deletion		48	439 668	210 382	High



Chr7:55,249,063
Amp3
Synonymous
A>G



Chr7:55,242,465 Amp2

Biological relevance

15-bp Inframe deletion

5 aa deleted -> Glu–Leu–Arg–Ala–Ala

EGFR mutated is constitutively active -> uncontrolled growth